JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON SATURDAY 04th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

1. For the function $f : [1, \infty) \to [1, \infty)$ defined by $f(x) = (x-1)^4 + 1$, among the two statements :

(I) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x)\}$ contains exactly two elements, and

(II) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x+1)\}$ is an empty set,

(1) only (I) is TRUE

(2) only (II) is TRUE

(3) both (I) and (II) are TRUE

(4) neither (I) nor (II) is TRUE

Sol. $f(x) = (x-1)^4 + 1$

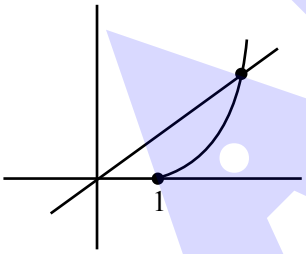
$f'(x) = 4(x-1)^3$; $f'(x) \geq 0$

$\therefore$ f(x) is increasing

$\therefore (x-1)^4 + 1 = x$

$(x-1)[(x-1)^3 - 1] = 0$

$x = 1, x = 2$ are two solutions



Now

$f^{-1}(x) = (x-1)^{1/4} + 1$

$f^{-1}(x+1) = x^{1/4} + 1$

$f^{-1}(x+1) = f(x)$

$x^{1/4} + 1 = (x-1)^4 + 1$

$x = (x-1)^{16}$

Above equation has one solution using graph.

2. Let $S = \{z \in \mathbb{C} : z^2 + 4z + 16 = 0\}$.

Then $\sum_{z \in S} |z + \sqrt{3}i|^2$ is equal to :

(1) 42

(2) 23

(3) 27

(4) 38

Sol. $z^2 + 4z + 16 = 0$

$\Rightarrow (z+2)^2 = -12$

$\Rightarrow z = -2 \pm 2\sqrt{3}i$

$\Rightarrow S = \{-2 + 2\sqrt{3}i, -2 - 2\sqrt{3}i\}$

$\therefore \sum_{z \in S} |z + \sqrt{3}i|^2$

$= |-2 + 2\sqrt{3}i + \sqrt{3}i|^2 + |-2 - 2\sqrt{3}i + \sqrt{3}i|^2$

$= |-2 + 3\sqrt{3}i|^2 + |-2 - \sqrt{3}i|^2$

$= 38$

3. If the system of equations :

$x + y + z = 5$

$x + 2y + 3z = 9$

$x + 3y + \lambda z = \mu$

has infinitely many solutions, then the value of $\lambda + \mu$ is :

(1) 16

(2) 18

(3) 19

(4) 21

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Sol. $D = \lambda - 5$

$D_1 = \lambda + \mu - 18$

$D_2 = 4\lambda - 2\mu + 6$

$D_3 = \mu - 13$

Here $x = \frac{D_1}{D}$; $y = \frac{D_2}{D}$; $z = \frac{D_3}{D}$

For infinitely many solutions

$D = D_1 = D_2 = D_3 = 0$

$\Rightarrow \lambda = 5$; $\mu = 13$

$\therefore \lambda + \mu = 18$

4. If $\alpha = 1$ and $\beta = 1 + i\sqrt{2}$, where $i = \sqrt{-1}$ are two roots of the equation

$x^3 + ax^2 + bx + c = 0$, a, b, $c \in \mathbb{R}$, then $\int_{-1}^{1} \left(x^3 + ax^2 + bx + c\right)dx$ is equal to :

(1) –2 (2) –4

(3) –8 (4) –10

Sol. Roots are $1 \pm i\sqrt{2}$ & 1

–a = sum of roots

a = –3

$-c = 1\left(1+\sqrt{2}\,i\right)\left(1-i\sqrt{2}\right)$

c = –3

$I = \int_{-1}^{1} \left(x^3 - 3x^2 + bx - 3\right)dx$

$= 2\int_{0}^{1} \left(-3x^2 - 3\right)dx$

$= 2(-x^3 - 3x)_0^1 = -8$

5. If the quadratic equation $(\lambda + 2)x^2 - 3\lambda x + 4\lambda = 0$, $\lambda \neq -2$, has two positive roots, then the number of possible integral values of λ is :

(1) 1 (2) 2

(3) 3 (4) 4

Sol. $f(x) = (\lambda + 2)x^2 - 3\lambda x + 4\lambda$

C-1 $af(0) > 0$

$(\lambda + 2)4\lambda > 0$

$\Rightarrow \lambda < -2$ or $\lambda > 0$

C-2 $-\frac{b}{2a} > 0$

$\frac{3\lambda}{2(\lambda+2)} > 0$

$\Rightarrow \lambda < -2$ or $\lambda > 0$

C-3 $D \geq 0$

$(-3\lambda)^2 - 4(\lambda + 2) \times 4\lambda \geq 0$

$\lambda(7\lambda + 32) \leq 0$

$\Rightarrow \lambda \in \left[\frac{-32}{7}, 0\right]$

Intersection of C-1, C-2 and C-3

$\Rightarrow \lambda \in \left[\frac{-32}{7}, -2\right)$

$\lambda \in [-4.57, -2)$

$\Rightarrow \lambda = -4, -3$

$\therefore$ Number of values of $\lambda = 2$

6. Let $A = \begin{bmatrix} 1 & 2 & 7 \\ 4 & -2 & 8 \\ 3 & 8 & -7 \end{bmatrix}$ and $\det(A - \alpha I) = 0$, where α is a real number. If the largest possible value of α is p, then the circle $(x-p)^2 + (y-2p)^2 = 320$, intersects the co-ordinate axes at

(1) 1 point

(2) 2 points

(3) 3 points

(4) 4 points

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Sol. $\begin{vmatrix} 1-\alpha & 2 & 7 \\ 4 & -2-\alpha & 8 \\ 3 & 8 & -7-\alpha \end{vmatrix} = 0$

$\Rightarrow (1-\alpha)[(\alpha+2)(\alpha+7)-64] - 2[-28-4\alpha-24] + 7[32+6+3\alpha] = 0$

$\Rightarrow \alpha^3 + 8\alpha^2 - 88\alpha - 320 = 0$

$(\alpha-8)(\alpha^2+16\alpha+40) = 0$

$\alpha = 8, \alpha = -8 \pm 2\sqrt{6}$

So $p = -8$

$\Rightarrow (x-8)^2 + (y-16)^2 = 320 = 8^2 + 16^2$

put $y = 0 \Rightarrow x = 16, 0$

put $x = 0 \Rightarrow y = 32, 0$

$\Rightarrow (16, 0), (0, 32), (0, 0)$

$\Rightarrow$ 3 points of intersection

7. Let $\alpha = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + ...\infty$ and $\beta = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + ...\infty$. Then the value of $(0.2)^{\log_{\sqrt{5}}(\alpha)} + (0.04)^{\log_5(\beta)}$ is equal to :

(1) 4 (2) 5

(3) 8 (4) 25

Sol. $\alpha = \frac{1/4}{1-\frac{1}{2}} = \frac{1}{2}$

$\beta = \frac{1/3}{1-\frac{1}{3}} = \frac{1}{2}$

$(0.2)^{\log_{\sqrt{5}}\alpha} = \alpha^{\log_{\sqrt{5}}\frac{1}{5}} = \left(\frac{1}{2}\right)^{-2} = 4$

$(0.04)^{\log_5\beta} = 5^{-2\log_5\beta} = \left(\frac{1}{2}\right)^{-2} = 4$

$\Rightarrow 4 + 4 = 8$

8. For 10 observations $x_1, x_2, ..., x_{10}$, if $\sum_{i=1}^{10}(x_i+2)^2 = 180$ and $\sum_{i=1}^{10}(x_i-1)^2 = 90$, then their standard deviation is :

(1) 2 (2) $\sqrt{3}$

(3) $2\sqrt{2}$ (4) 3

Sol. $\sum_{i=1}^{10}(x_i+2)^2 = 180$

$\sum_{i=1}^{10}x_i^2 + 4\sum_{i=1}^{10}x_i + \sum_{i=1}^{10}4 = 180$

$\sum_{i=1}^{10}x_i^2 + 4\sum_{i=1}^{10}x_i = 180 - 40 \quad ...(1)$

Also $\sum_{i=1}^{10}(x_i-1)^2 = 90$

$\sum_{i=1}^{10}x_i^2 - 2\sum_{i=1}^{10}x_i + \sum_{i=1}^{10}1 = 90$

$\sum_{i=1}^{10}x_i^2 - 2\sum_{i=1}^{10}x_i = 90 - 10 \quad ...(2)$

$\sum_{i=1}^{10}x_i^2 + 4\sum_{i=1}^{10}x_i = 140$

$\sum_{i=1}^{10}x_i^2 - 2\sum_{i=1}^{10}x_i = 80$

$\sum_{i=1}^{10}x_i^2 = 100$ and $\sum_{i=1}^{10}x_i = 10$

$\sigma^2 = \frac{\sum_{i=1}^{10}x_i^2}{N} - \left(\frac{\sum_{i=1}^{10}x_i}{N}\right)^2$

$= \frac{100}{10} - \left(\frac{10}{10}\right)^2$

$\sigma^2 = 10 - 1 = 9$

$\Rightarrow \sigma = 3$

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9. In the expansion of $\left(9x-\frac{1}{3\sqrt{x}}\right)^{18}, x>0$, if the term independent of x is (221)k. then k is equal to:

(1) 84 (2) 78

(3) 168 (4) 198

Sol. General term in above expression is

$$T_{r+1}={}^{18}C_r(9x)^{18-r}\left(-\frac{1}{3\sqrt{x}}\right)^r$$

$$=\left(-\frac{1}{3}\right)^r {}^{18}C_r 9^{18-r}\cdot x^{18-\frac{3r}{2}}$$

For term independent of x, $18-\frac{3r}{2}=0\Rightarrow r=12$

$\therefore$ Coefficient of term independent of x

$$=\left(-\frac{1}{3}\right)^{12} {}^{18}C_{12}9^{18-12}$$

= 18564

= 221 k (given)

$\Rightarrow$ k = 84

10. Let P(3cos α, 2sin α), $\alpha\neq 0$, be a point on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, Q be a point on the circle $x^2+y^2-14x-14y+82=0$ and R be a point on the line x + y = 5 such that the centroid of the triangle PQR is $\left(2+\cos\alpha, 3+\frac{2}{3}\sin\alpha\right)$. Then the sum of the ordinates of all possible points R is:

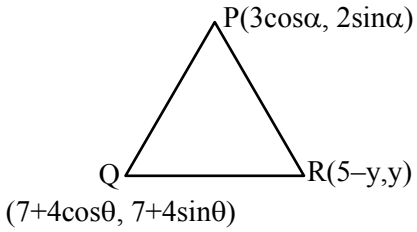
(1) 6 (2) 2

(3) 4 (4) 8

Sol. Centroid :

$$\left(\frac{3\cos\alpha+7+4\cos\theta+5-y}{3}, \frac{2\sin\alpha+7+4\sin\theta+y}{3}\right)$$

$$=\left(\cos\alpha+2, \frac{2}{3}\sin\alpha+3\right)$$



On comparison

$$\cos\alpha=\frac{y-6}{4} \text{ \& } \sin\alpha=\frac{2-y}{4}$$

$\sin^2\alpha+\cos^2\alpha=1\Rightarrow (y-6)^2+(y-2)^2=16$

$y^2-8y+12=0$

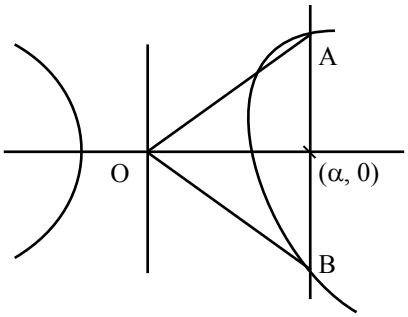
$\Rightarrow$ Sum of ordinates = 8

11. Let $H:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be a hyperbola such that the distance between its foci is 6 and the distance between its directrices is $\frac{8}{3}$. If the line x = α intersects the hyperbola H at the points A and B such that the area of the triangle AOB is $4\sqrt{15}$, where O is the origin, then α^2 equals

(1) 12 (2) 16

(3) 24 (4) 25

Sol. $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$



Given:

2ae = 6 $\Rightarrow$ ae = 3

$$\frac{2a}{e}=\frac{8}{3}\Rightarrow\frac{a}{e}=\frac{4}{3}$$

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$a^2 = 4 \quad e = \frac{3}{2} \quad b^2 = 5$

$\frac{x^2}{4} - \frac{y^2}{5} = 1$ solve with $x = \alpha$

$y = \pm\sqrt{\frac{5(\alpha^2-4)}{4}}$

$A\left(\alpha, \sqrt{\frac{5(\alpha^2-4)}{4}}\right) \quad B\left(\alpha, -\sqrt{\frac{5(\alpha^2-4)}{4}}\right)$

Ar. of $\Delta OAB = \frac{1}{2}|\alpha|\sqrt{5(\alpha^2-4)} = 4\sqrt{15}$

$\Rightarrow 16\times 15 = \frac{1}{4}\alpha^2(5)(\alpha^2-4)$

$\Rightarrow \alpha^2 = 16$

12. $\max\limits_{0 \le x \le \pi}\left(16\sin\left(\frac{x}{2}\right)\cos^3\left(\frac{x}{2}\right)\right)$ is equal to :

(1) $\frac{3\sqrt{3}}{2}$ (2) $3\sqrt{3}$

(3) $4\sqrt{3}$ (4) $6\sqrt{3}$

Sol. $E = 16\sin\frac{x}{2}\cos^3\frac{x}{2}$

$E = 4\sin x\,[1+\cos x]$

$\frac{dE}{dx} = 4\,[\cos x + \cos 2x]$

$= 8\cos\frac{3x}{2}\cos\frac{x}{2} = 0$

$\Rightarrow \cos\frac{3x}{2} = 0$ or $\cos\frac{x}{2} = 0$

$\Rightarrow x = \left\{\frac{\pi}{3}, \pi\right\}$ are critical points of the function

$E(0) = 0$

$E(\pi) = 0$

$E\left(\frac{\pi}{3}\right) = 3\sqrt{3}$

$\therefore$ maximum value of $E = 3\sqrt{3}$

13. The shortest distance between the lines

$\vec{r} = \left(\frac{1}{3}\hat{i} + 2\hat{j} + \frac{8}{3}\hat{k}\right) + \lambda\left(2\hat{i} - 5\hat{j} + 6\hat{k}\right)$

and $\vec{r} = \left(-\frac{2}{3}\hat{i} - \frac{1}{3}\hat{k}\right) + \mu\left(\hat{j} - \hat{k}\right)$, $\lambda, \mu \in \mathbb{R}$, is :

(1) $\sqrt{5}$ (2) 3

(3) $2\sqrt{3}$ (4) $\sqrt{15}$

Sol. $\vec{a}_1 = \frac{1}{3}\hat{i} + 2\hat{j} + \frac{8}{3}\hat{k}$

$\vec{a}_2 = \frac{-2}{3}\hat{i} - \frac{1}{3}\hat{k}$

$\vec{b}_1 = 2\hat{i} - 5\hat{j} + 6\hat{k}, \quad \vec{b}_2 = \hat{j} - \hat{k}$

$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -5 & 6 \\ 0 & 1 & -1 \end{vmatrix} = -\hat{i} + 2\hat{j} + 2\hat{k}$

$\vec{a}_1 - \vec{a}_2 = \hat{i} + 2\hat{j} + 3\hat{k}$

$S.D = \left|\frac{(\vec{a}_1 - \vec{a}_2)\cdot(\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}\right|$

$S.D = \left|\frac{(\hat{i} + 2\hat{j} + 3\hat{k})\cdot(-\hat{i} + 2\hat{j} + 2\hat{k})}{3}\right|$

$= \frac{9}{3} = 3$

14. If $\left(2\alpha + 1, \alpha^2 - 3\alpha, \frac{\alpha - 1}{2}\right)$ is the image of $(\alpha, 2\alpha, 1)$ in the line $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z}{1}$, then the possible value(s) of α is(are) :

(1) Only 3

(2) Only 3 and –1

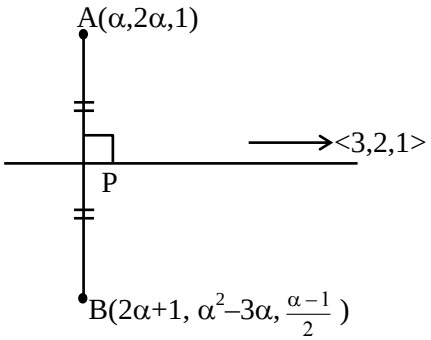
(3) Only 3, $\frac{1}{4}$ and –1

(4) Only 3 and $\frac{1}{4}$

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Sol. $\therefore$ P is mid point of AB and lies on the given line

$$\Rightarrow P\left(\frac{3\alpha+1}{2}, \frac{\alpha^2-\alpha}{2}, \frac{\alpha+1}{4}\right)$$



Now put coordinates of P in the line

$$\Rightarrow \frac{\frac{3\alpha+1}{2}-2}{3} = \frac{\frac{\alpha^2-\alpha}{2}-1}{2} = \frac{\alpha+1}{4}$$

$$\Rightarrow \frac{\alpha-1}{2} = \frac{\alpha+1}{4} = \frac{\alpha^2-\alpha-2}{4}$$

$$\Rightarrow \alpha = 3$$

Now for $\alpha = 3$, clearly AB is perpendicular to $3\hat{i}+2\hat{j}+k$

for $\alpha = 3$ point B is image of point A

15. Let $\hat{u}$ and $\hat{v}$ be unit vectors inclined at an acute angle such that $|\hat{u}\times\hat{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \lambda\hat{u}+\hat{v}+(\hat{u}\times\hat{v})$. Then λ is equal to :

(1) $\frac{4}{3}(\vec{A}.\hat{u}) - \frac{2}{3}(\vec{A}.\hat{v})$

(2) $\frac{2}{3}(\vec{A}.\hat{u}) - \frac{1}{3}(\vec{A}.\hat{v})$

(3) $\frac{4}{3}(\vec{A}.\hat{u}) + \frac{2}{3}(\vec{A}.\hat{v})$

(4) $(\vec{A}.\hat{u}) - \frac{1}{2}(\vec{A}.\hat{v})$

Sol. $|\hat{u}\times\hat{v}| = \frac{\sqrt{3}}{2}$

$$|\hat{u}||\hat{v}|\sin\theta = \frac{\sqrt{3}}{2} \quad \therefore \theta = \frac{\pi}{3}$$

and $\hat{u}\cdot\hat{v} = |\hat{u}||\hat{v}|\cos\frac{\pi}{3} = \frac{1}{2}$

$$\vec{A} = \lambda\hat{u}+\hat{v}+\hat{u}\times\hat{v} \quad \ldots(1)$$

Dot with $\hat{u}$

$$\vec{A}\cdot\hat{u} = \lambda(1)+\hat{u}.\hat{v}+\hat{u}\cdot(\hat{u}\times\hat{v})$$

$$\vec{A}\cdot\hat{u} = \lambda+\frac{1}{2}$$

$$\Rightarrow 2\vec{A}\cdot\hat{u} = 2\lambda+1 \quad \ldots(2)$$

Dot equation (1) with $\hat{v}$

$$\vec{A}\cdot\hat{v} = \lambda(\hat{u}\cdot\hat{v})+\hat{v}\cdot\hat{v}+\hat{v}\cdot(\hat{u}\times\hat{v})$$

$$\vec{A}\cdot\hat{v} = \frac{\lambda}{2}+1$$

$$\vec{A}\cdot\hat{v} - \frac{\lambda}{2} = 1 \quad \ldots(3)$$

From (2) and (3)

$$2\vec{A}\cdot\hat{u} - 2\lambda = \vec{A}.\hat{v} - \frac{\lambda}{2}$$

$$\therefore \lambda = \frac{4}{3}\vec{A}\cdot\hat{u} - \frac{2}{3}\vec{A}.\hat{v}$$

16. Let for some $\alpha \in \mathbb{R}$, $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying $f(x+y) = f(x) + 2y^2 + y + \alpha xy$ for all $x, y \in \mathbb{R}$. If $f(0) = -1$ and $f(1) = 2$, then the value of $\sum_{n=1}^{5}(\alpha + f(n))$ is :

(1) 110 (2) 140

(3) 150 (4) 170

Sol. $f(x+y) = f(x) + 2y^2 + y + \alpha xy$ ...(1)

Put $x = 0$ in eq.(1)

$$\Rightarrow f(y) = -1 + 2y^2 + y$$

Now put $x = y = 1$ in eq. (1)

$$\Rightarrow f(2) = f(1) + 3 + \alpha$$

$$\Rightarrow 9 = 2 + 3 + \alpha \Rightarrow \alpha = 4$$

$$\text{Now } \sum_{n=1}^{5}(\alpha + f(n)) = \sum_{n=1}^{5}(2y^2 + y + 3)$$

$$= \frac{2\times5\times6\times11}{6} + \frac{5\times6}{2} + 3\times5 = 140$$

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17. Let A = {(a, b, c) : a, b, c are non-negative integers and a + b + 2c = 22}.

The n(A) is equal to :

(1) 121 (2) 124

(3) 144 (4) 169

Sol. $c = 0 \Rightarrow a + b = 22 \rightarrow {}^{22+2-1}C_{2-1} = {}^{23}C_1$

$c = 1 \Rightarrow a + b = 20 \rightarrow {}^{20+2-1}C_{2-1} = {}^{21}C_1$

$\vdots \quad \vdots \quad \vdots$

$c = 10 \Rightarrow a + b = 2 \rightarrow {}^{2+2-1}C_{2-1} = {}^{3}C_1$

$c = 11 \Rightarrow a + b = 0 \rightarrow {}^{0+2-1}C_{2-1} = {}^{1}C_1$

Number of non-negative integral solutions

${}^1C_1 + {}^3C_1 + {}^5C_1 + \ldots + {}^{23}C_1$

$1 + 3 + 5 \ldots\ldots + 23$

$= 144$

18. The area of the region bounded by the curves $x + 3y^2 = 0$ and $x + 4y^2 = 1$ is equal to :

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{4}{3}$ (4) $\frac{5}{3}$

Sol. $x = -3y^2$ and $x = 1 - 4y^2$

$$A = \int_{-1}^{1} \left((1-4y^2) + 3y^2\right) dy$$

$$A = \int_{-1}^{1} (1-y^2) dy = 2\int_{0}^{1} (1-y^2) dy$$

$$A = \frac{4}{3}$$

19. Let y = y(x) be the solution of the differential equation :

$$\frac{dy}{dx} + \left(\frac{6x^2 + \left(3x^2 + 2x^3 + 4\right)e^{-2x}}{\left(x^3+2\right)\left(2+e^{-2x}\right)}\right) y = 2 + e^{-2x},$$

$x \in (-1, 2)$, satisfying $y(0) = \frac{3}{2}$.

If $y(1) = \alpha(2 + e^{-2})$, than α is equal to :

(1) $\frac{13}{8}$ (2) $\frac{6}{13}$

(3) $\frac{12}{13}$ (4) $\frac{13}{12}$

Sol. $\text{I.F.} = e^{\int \frac{6x^2 + e^{-2x}(3x^2+2x^3+4)}{(x^3+2)(2+e^{-2x})} dx}$

$$= e^{\int \frac{6x^2 + e^{-2x}(3x^2-2x^3-4)+(4x^3+8)e^{-2x}}{(x^3+2)(2+e^{-2x})} dx}$$

Let $(x^3 + 2)(2 + e^{-2x}) = t$

$(6x^2 + e^{-2x}(3x^2 - 2x^3 - 4))dx = dt$

$$\text{I.F.} = e^{\ell n\left|(x^3+2)(2+e^{-2x})\right| - \int \frac{4e^{-2x}dx}{2+e^{-2x}}}$$

$$= e^{\ell n\left|(x^3+2)(2+e^{-2x})\right| - 2\ell n\left|2+e^{-2x}\right|}$$

$$= e^{\ell n\left|\frac{x^3+2}{2+e^{-2x}}\right|} = \frac{x^3+2}{2+e^{-2x}}$$

$$\Rightarrow y.\left(\frac{x^3+2}{2+e^{-2x}}\right) = \int \left(\frac{x^3+2}{2+e^{-2x}}\right)(e^{-2x} + 2) dx + C$$

$$y.\left(\frac{x^3+2}{2+e^{-2x}}\right) = \frac{x^4}{4} + 2x + C \text{ at } y(0) = \frac{3}{2} \Rightarrow C = 1$$

$$\text{So } y.\left(\frac{x^3+2}{2+e^{-2x}}\right) = \frac{x^4}{4} + 2x + 1$$

at x = 1

$$y.\left(\frac{3}{2+e^{-2}}\right) = \frac{1}{4} + 2 + 1$$

$$\Rightarrow y = \frac{13}{12}(e^{-2} + 2)$$

$$\text{So } \alpha = \frac{13}{12}$$

20. The integral $\int_0^1 \cot^{-1}\left(1 + x + x^2\right) dx$ is equal to :

(1) $2\tan^{-1}2 + \frac{1}{2}\log_e\left(\frac{5}{4}\right) + \frac{\pi}{2}$

(2) $2\tan^{-1}2 + \frac{1}{2}\log_e\left(\frac{5}{4}\right) - \frac{\pi}{2}$

(3) $2\tan^{-1}2 - \frac{1}{2}\log_e\left(\frac{5}{4}\right) + \frac{\pi}{2}$

(4) $2\tan^{-1}2 - \frac{1}{2}\log_e\left(\frac{5}{4}\right) - \frac{\pi}{2}$

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Sol. $= \int_0^1 \tan^{-1}\left(\frac{1}{1+x+x^2}\right)dx$

$= \int_0^1 \tan^{-1}\left(\frac{(x+1)-x}{1+x(x+1)}\right)dx$

$= \int_0^1 (\tan^{-1}(x+1)-\tan^{-1}x)dx$

$= \left(x\tan^{-1}(x+1)-\frac{1}{2}\ell n|1+(1+x)^2|+\tan^{-1}(x+1)\right)_0^1$

$-\left(x\tan^{-1}x-\frac{1}{2}\ell n|1+x^2|\right)_0^1$

$= \left(2\tan^{-1}2-\frac{\pi}{4}-\frac{1}{2}\ell n\frac{5}{2}\right)-\left(\frac{\pi}{4}-\frac{1}{2}\ell n2\right)$

$= 2\tan^{-1}2-\frac{1}{2}\ell n\left(\frac{5}{4}\right)-\frac{\pi}{2}$

SECTION-B

21. From a month of 31 days, 3 different dates are selected at random. If the probability that these dates are in an increasing A.P. is equal to $\frac{a}{b}$, where a, b $\in$ N and *gcd*(a,b) = 1, then a + b is equal to _____.

Sol. Total numbers of ways of selecting

3 numbers $= {}^{31}C_3 = \frac{31\times30\times29}{3\times2\times1} = 4495$

a, b, c are in A.P.

$\Rightarrow$ either a & c both odd numbers

or both even numbers

$= {}^{16}C_2 + {}^{15}C_2$

$= 120 + 105$

$= 225$

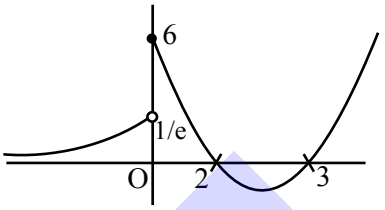
Probability $= \frac{225}{4495} = \frac{45}{899}$

$= \frac{a}{b} \Rightarrow a+b = 944$

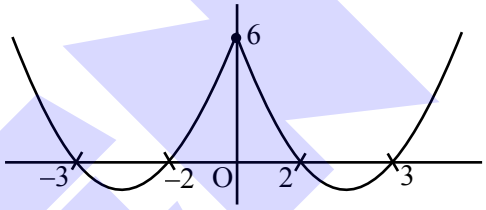
22. Let $f(x) = \begin{cases} e^{x-1} & ,x<0 \\ x^2-5x+6 & ,x\geq 0 \end{cases}$ and

$g(x) = f(|x|) + |f(x)|$. If the number of points where g is not continuous and is not differentiable are α and β respectively, then $\alpha+\beta$ is equal to _____

Sol. Graph of f(x)

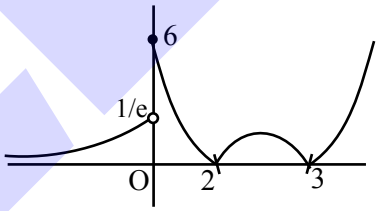


Now graph of f(|x|)



f(|x|) is continuous function and it is non diff. at x = 0

Graph of |f(x)|



|f(x)| is discontinuous at x = 0

|f(x)| is non-diff. at x = 0, 2, 3

g(x) = f(|x|) + |f(x)|

g(x) will be discontinuous at x = 0

g(x) will be non diff at x = 0, 2, 3

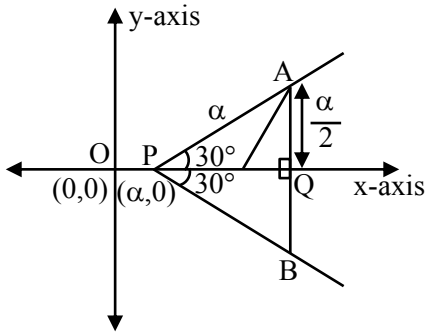
$\alpha = 1, \beta = 3$

$\alpha+\beta = 1+3 = 4$

23. Let A, B be points on the two half-lines $x-\sqrt{3}|y| = \alpha$, $\alpha > 0$ at a distance of α from their point of intersection P. The line segment AB meets the angle bisector of the given half-lines at the point Q. If PQ = $\frac{9}{2}$ and R is the radius of the circumcircle of ΔPAB, then $\frac{\alpha^2}{R}$ is equal to _____.

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Sol. ΔPAB is equilateral



$$PQ = \frac{\sqrt{3}\alpha}{2} = \frac{9}{2} \qquad \therefore \quad \alpha = \frac{9}{\sqrt{3}}$$

$$\alpha = 3\sqrt{3}$$

$$\frac{\alpha}{2r} = \cos 30°$$

$$\alpha = \frac{\sqrt{3}}{2} \cdot 2R$$

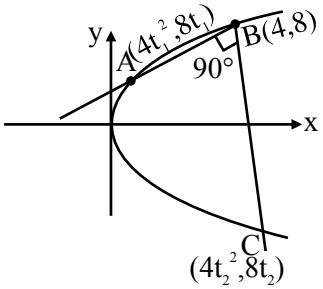
$$\alpha = \sqrt{3}\, R$$

$$\boxed{R = 3}$$

$$\frac{\alpha^2}{R} = \frac{(3\sqrt{3})^2}{3} = 9$$

24. Let A, B and C be the vertices of a variable right angled triangle inscribed in the parabola $y^2 = 16x$. Let the vertex B containing the right angle be (4, 8) and the locus of the centroid of ΔABC be a conic C_0. Then three times the length of latus rectum of C_0 is _____.

Sol.



Variable point A & C are shown in the figure

$\because m_{AB} . m_{BC} = -1$

$\Rightarrow t_1 + t_2 + t_1 t_2 = -5 \ldots(1)$

Suppose locus of centroid of ΔABC is (h,k)

$\therefore 3h = 4 + 4t_1^2 + 4t_2^2$ & $3k = 8 + 8t_1 + 8t_2$

by eliminating t_1 & t_2 using equation (1) also

we get $h = \frac{9}{48}k^2 + \frac{40}{3}$

$\therefore$ locus of centroid of parabola is

$$x = \frac{9}{48}y^2 + \frac{40}{3}$$

$$\therefore \ \ell(\text{L.R.}) = \frac{48}{9}$$

$$\therefore \ 3\ell(\text{L.R.}) = \frac{48}{3} = 16$$

25. Let f be a twice differentiable function such that

$$f(x) = \int_0^x \tan(t - x)\,dt - \int_0^x f(t)\tan t\,dt, \ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Then $f''\left(\frac{\pi}{6}\right) + 12f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right)$

is equal to ____.

Sol. $f(x) = \int_0^x \tan(x - t - x)\,dt - \int_0^x f(t)\tan t\,dt$

$f(x) = -\int_0^x \tan t\,dt - \int_0^x f(t)\tan t\,dt$

$f'(x) = -\tan x - f(x)\tan x$

$\frac{dy}{y+1} = -\tan x\,dx$

$y + 1 = \cos x . c$

$f(0) = 0 \Rightarrow 1 = c$

$y = \cos x - 1 = f(x)$

$f\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} - 1$

$f'(x) = -\sin x$

$f'\left(\frac{-\pi}{6}\right) = \frac{1}{2}$

$f''(x) = -\cos x$

$f''\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$

$f''\left(\frac{\pi}{6}\right) + 12f'\left(\frac{-\pi}{6}\right) + f\left(\frac{\pi}{6}\right) = 5$

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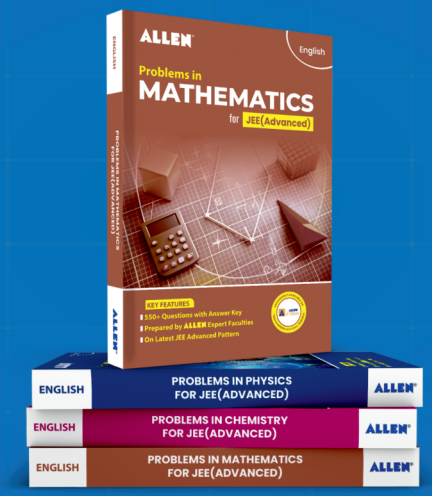


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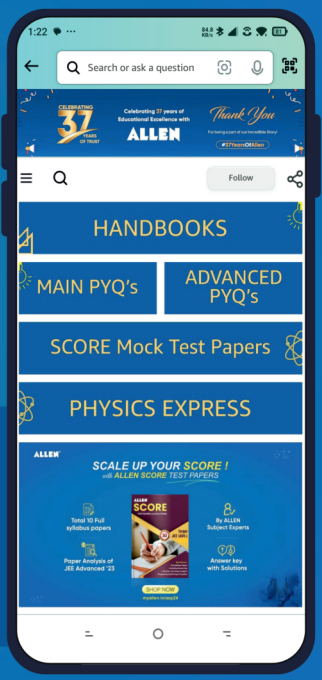
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